Teryx EV system — 4-string dual-Teensy (CAN_LAYOUT_V2), wire numbers from docs/PINMAP.yaml (generated)

HIGH VOLTAGE — isolated, B− never to chassis (LATER, Sparky)

String 1 (A, CAN1)
String 2 (A, CAN2)
String 3 (B, CAN1)
String 4 (B, CAN2)
H01
4× Mersen A6T200 200 A Class T on insulated standoffs
SERVICE DISCONNECT = pull all 4 fuses (no MSD)
H05
HV fuse DC-DC in 15–20 A
≥250 V DC (confirm)
H12
Common B+ bus
H06
EV200 B+
H08
Precharge relay AEV14012
(vs TE EV11 — CONFLICT)
P03
50 Ω 100 W resistor
P01
H07
Kelly KLS8080 NPS
168 V 550 A sin/cos
H09
H13
Zero 75-7R (used)
sin/cos · spin DONE
H10
EV200 B−
S05
S25
E02
H11
H12
LEM DHAB A (S1+S2) / B (S3+S4)
on string B− leads
H10
Elcon TDC-JH-144-12 DC-DC
(isolation barrier)
Elcon charger
H13
H14
NACS inlet (AC)
F-ELC Mersen A6T60 60 A
6 AWG · holder TBD
E03
A:24 RESERVED spare (room 2026-10-23)
Precharge/weld = Kelly CAN B+ vs BICM sum
AMC1311EVM only if bench gate fails
E03

LOW VOLTAGE — 12 V− bonded to chassis

12 V bus bar
P02
F-MAIN 12 V (size TBD)
P03
P04
P05
P06
A01
F-KEL Kelly PWR
F-COIL coil feed
F-EBOX electronics
F-TAB tablet
F-AUX aux loads
P07
F-BUCK 3 A mini blade
LATER — reserved: MOTOR_FAN
MCP23017 bit 14 → IRLB8721 + 1N4007 → fan relay
LATER — reserved: BUZZER
MCP23017 bit 13 → 12 V buzzer
P07
5 V buck (set 5.00 V)
Zener 1N5339B
P08
P09
5 V terminal block
GND terminal block (single point)
P12
G04
P08
G06
G05
P11
Teensy A (dual_a) — master
CAN1 22/23 S1 · CAN2 0/1 S2 · CAN3 30/31 veh
Serial4 TX17 → ESP32 · pins 11/12/21 coils
E01
E01
S01
S02
Teensy B (dual_b) — watchdog
CAN1 S3 · CAN2 S4 · CAN3 veh (stub)
e-stop input pin UNDEFINED
S30
ESP32 gateway (read-only)
GPIO16 RX · GPIO17 UNUSED
X01
A01
iPad (Wi-Fi AP)
C01
C03
C10
C05
P10
C02
C04
G07
S01
C06
E03
SN65 #1
SN65 #2
K01
SN65 #5 +120 Ω
SN65 #3
SN65 #4
E09
S12
S10
C10
S03
C11
C12
S16
E09
S17
SN65 #6 (stub)
Coil FETs + 10k gate pull-downs
K03
BICM X2 harness ×4: 9=5V 10=GND 11=CAN_L 12=CAN_H
(CAN H/L unconfirmed)
S20
S50
5139 pull-up
Throttle clamp (MCP4131 + 4066 + 74HC123)
S14
E-stop NC block 1 (11/12)
EV200 coils
E-stop NC block 2 (11/12)
precharge + KSI
E-stop NC block 3 (11/12)
→ Teensy B (pin UNDEFINED)
Park switch 3-pole
Pedal (Hall)
Low-cap relay → Kelly ECO
Aux relay 30/87/85/86
A02
NO aux button
A04
Horn / lights

Red = unconfirmed / UNDEFINED. Grey dashed = LATER, reserved (no logic). Labels = wire numbers from docs/PINMAP.yaml